Check for updates

Received: 22 November 2021 | Revised: 5 February 2024 | Accepted: 19 February 2024

DOI: 10.1111/jbfa.12794

JBFA

ARTICLE

The unintended cost of data breach notification laws: Evidence from managerial bad news hoarding

Ivan Obaydin | **Limin Xu** | **Ralf Zurbruegg**

Business School, University of Adelaide, Adelaide, South Australia, Australia

Correspondence
Ralf Zurbruegg, Business School, University of Adelaide, Adelaide 5005, SA, Australia.
Email: ralf.zurbrugg@adelaide.edu.au

Abstract

We investigate how nonfinancial disclosure laws exacerbate agency issues in firms. Our analysis focuses on the staggered adoption of state-level notification laws that require firms to disclose data breaches. Our findings reveal that the introduction of these laws increases the risk of stock price crashes. Managers appear motivated to accumulate unfavorable news in an effort to prevent market overreactions associated with the mandatory disclosure of data breaches. Cross-sectional analyses also reveal that the impact is stronger when managers have a greater incentive, or greater ability, to hoard information. Our results highlight that nonfinancial disclosures can have unintended consequences for firm information asymmetries and potentially adverse market impacts in cases where the regulation is unable to consider all stakeholders.

KEYWORDS
bad news hoarding, cybersecurity risk, data breach notification laws, managerial incentives, mandatory nonfinancial disclosure laws, stock price crash risk

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.
© 2024 The Authors. *Journal of Business Finance & Accounting* published by John Wiley & Sons Ltd.

1 | INTRODUCTION

Mandatory disclosure regimes are often used to provide consumer protection as an alternative to direct regulation. Although this type of legislation is designed with the consumer in mind, it can also affect other stakeholders. Unlike mandatory financial disclosures that can reduce information asymmetry between a firm's managers and investors (Greenstone et al., 2006; Li & Yang, 2016), consumer protection laws are not necessarily meant to minimize these agency issues, potentially leading to several unintended consequences. This motivates us to investigate if and how agency problems are affected by a set of recently enacted state-level data breach notification (DBN) laws—laws that are now adopted in every US state.

When personal consumer information is compromised due to a data breach, the costs to firms can be substantial. Kamiya et al. (2021) documented a 0.84% decline in cumulative abnormal returns (CARs) around breach announcements for publicly listed firms, whereas IBM estimates that the average cost of a data breach in the United States is in excess of $8 million.[1] Data breaches also impose indirect costs like increased loan spreads (Huang & Wang, 2021), reputation loss (Kamiya et al., 2021), and customer attrition (Ablon et al., 2016). These breaches can occur for many reasons, but with firms moving to rely more heavily on electronic data storage, cyber security risks, in particular, have risen.

As a response to demands for improved consumer protection, US states have enacted data breach disclosure legislation. From the date of their enactment, breached companies are required by law to notify individuals whose personal information has been compromised. California became the first state to adopt a DBN law in 2003, and by 2018, all US states had mandatory data breach disclosure laws requiring firms to inform affected individuals so that they can take remedial measures to limit their exposure to, for example, identity theft. Although these laws do not necessarily require firms to make public announcements, data breaches inevitably become public information if for no other reason than the media learning of the events from the affected parties.[2]

Prior to the enactment of DBN laws, firms had the option to withhold information surrounding a data breach. As Amir et al. (2018) highlighted, managers have an incentive to withhold bad news surrounding data breaches, leading to cyberattacks being underreported and difficult for investors to discover. With that choice removed by mandatory disclosure legislation, firms now risk both reputational damage and stock price slumps. Though investing in security can help minimize the risk of a data breach (Murciano-Goroff, 2019), the payoff may not be cost-effective as it is nearly impossible to altogether remove the risk of experiencing a breach.[3] Consequently, managers might pursue other strategies to minimize negative impacts, like timing the release of other bad news, given that disclosures clustered together can compound market overreactions (Mian & Sankaraguruswamy, 2012).

Timing the release of bad news is not uncommon. A good example is the Volkswagen emissions scandal in 2015, where the company admitted to cheating on emission tests for its diesel cars. The scandal resulted in a decline in Volkswagen's reputation. After the scandal, it was revealed that Volkswagen had delayed releasing news about financial losses and management changes in an attempt to further mitigate the additional negative impact on its reputation and stock price.[4]

Similarly, even in the absence of an actual data breach, companies with a high probability of experiencing a cyberattack might be concerned about any other information they may have to release under a mandatory disclosure of a potential future breach. A string of negative news can significantly impact market sentiment, leading to a pessimistic

[1] This applies to all types of businesses, not just publicly listed firms (see *Cost of a Data Breach Report 2020* by IBM Security at https://www.ibm.com/security/digital-assets/cost-data-breach-report/#/).

[2] It is also common for data breach notification laws to require the affected firm to report breaches to the State Attorney-General's office and/or other state bodies.

[3] Even the most technologically advanced companies are subject to cybersecurity breaches. In the first half of 2021, Facebook, Instagram, LinkedIn, Pixlr and Microsoft Exchange all experienced cyberattacks that comprised customer security. See https://www.identityforce.com/blog/2021-data-breaches for a list of recent data breaches.

[4] See: https://www.nytimes.com/2016/02/06/business/international/volkswagen-earnings-emissions.html.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

outlook. This can create a bearish sentiment, prompting investors to engage in widespread selling. For instance, Mian and Sankaraguruswamy (2012) showed that stock prices exhibit more pronounced negative reactions to unfavorable news during periods of bearish sentiment when compared to those marked by bullish sentiment. Furthermore, investors can exhibit a confirmation bias, wherein they tend to give greater consideration to information that aligns with their preexisting beliefs or expectations. When a series of adverse news events unfolds one after the other, it intensifies the notion that a company is facing severe issues. As a result, investors might react more strongly to validate their existing convictions, potentially resulting in overreactions.

Supporting the above point, Gennaioli et al. (2015) argued that a persistent stream of negative news can trigger a profound shift in investor beliefs, potentially contributing to a financial crisis. Therefore, to prevent facing a sequence of negative news, managers are incentivized to delay the disclosure of adverse information following a data breach in order to protect their corporate reputation and stock price. This perspective is supported by Xu et al. (2019), who showed that companies, after experiencing a data breach, often opt to postpone the release of information regarding the incident. They also resort to earnings management practices as a strategy to obscure less favorable short-term financial performance. Although there may be multiple incentives for managers to delay the release of negative information (Hutton et al., 2009; Jin & Myers, 2006; Kim et al., 2011; Kothari et al., 2009), a consequence of such actions is that they lead to increased stock price crash risk. When the accumulation of negative information becomes impossible to hide, it can trigger a stock price crash, resulting in a pronounced left tail in the return distribution (Cao et al., 2019; Hanlon et al., 2023; Karamanou et al., 2021).

To test the hypothesis that crash risk increases following the adoption of DBN laws, we employ a difference-in-differences (DiD) approach that exploits the exogenous variation arising from the staggered enactment of these laws across different states. The staggered implementation of these laws creates a plausible natural experiment to examine how they influence firm preferences regarding the hoarding of negative news and its impact on crash risk. Our dataset comprises 37,987 firm-year observations of US publicly traded companies spanning from 2001 to 2019.

Our empirical investigation first focuses on whether there is a significant effect of DBN laws on the capital market by checking for evidence that their introduction changes firm crash risk. We proxy for firm-specific crash risk using three measures common in the literature: the negative skewness coefficient of firm-specific returns (*NSKEW*), the down-to-up volatility of firm-specific returns (*DUVOL*), and a count (*COUNT*) of the difference in the number of days with negative, as opposed to positive, extreme firm-specific returns (Chen et al., 2001; Hutton et al., 2009; Jin & Myers, 2006). Our results show a significant rise in crash risk among firms affected by the enactment of DBN laws. These results hold even when we include a comprehensive set of firm-level control variables alongside year and firm fixed-effects. These results are also economically significant. The adoption of DBN laws leads to a rise in crash risk by a minimum of 5% across all three crash risk measures, relative to their respective standard deviations. Addressing potential measurement error concerns, we find no evidence of a pre-trend, indicating that the parallel trends assumption is upheld. We also conduct permutation tests to ensure that our results could not be replicated through a random draw.

We then turn our attention to examine the channel by which crash risk rises following the enactment of DBN laws. We find evidence consistent with our hypothesis that there is a rise in the withholding of negative news after states enact DBN laws. Using Bao et al.'s (2019) measure of residual short interest, which proxies for managerial private bad news hoarding, we find that it increases by 5.2% following DBN legislation, relative to its standard deviation. In addition, we find a significant rise in the difference between optimistic and pessimistic management earnings guidance issued (a 7.7% rise relative to its standard deviation). Supporting this result, we also find that firms increase the number of voluntary 8K filings that they post, with this rise primarily driven by the release of positive, rather than negative, news.

In cross-sectional analyses, we find that firms in states that have enacted DBN laws that allow for private causes of action are more likely to be affected, likely because this exposes them to greater litigation risk. Moreover, firms with lesser information security awareness, or those that have a weaker information environment or are in industries that are more exposed to data breaches, all have greater crash risk.

Finally, we conduct additional robustness and sensitivity tests. As Baker et al. (2022) highlighted that staggered DiD estimates can sometimes be biased due to a combination of the staggered treatment timing and possible treatment

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

effect heterogeneity present within the data, we follow their advice and present results using a stacked regression approach. Moreover, we show that our results hold when accounting for firms that have operations in multiple states, when we exclude the global financial crisis period, and when we include higher level fixed effects.

This study contributes to the limited literature on the corporate effects of nonfinancial disclosure regulation (Leuz & Wysocki, 2016). We show that DBN laws can inadvertently increase agency problems by worsening financial transparency, as improved operational transparency incentivizes managers to hoard other bad news. Our findings suggest that consumer protection laws can have unintended consequences by overlooking potential side effects for non-targeted stakeholders.

We also contribute to the growing literature that examines the impact of data breaches (Garg, 2020; Huang & Wang, 2021; Kamiya et al., 2021) and DBN laws (Ashraf & Sunder, 2023; Boasiako & Keefe, 2021; Huang & Wang, 2021; Liu & Ni, 2024) on firm behavior. In particular, by showing that these laws can worsen the information asymmetry between managers and investors, thereby increasing stock price tail risk for shareholders.

Finally, our research contributes to the stock price crash risk literature by discovering how nonfinancial disclosure regimes can cause such crashes. Prior research has shown that crash risk is linked to corporate transparency and governance (Hong et al., 2017; Hutton et al., 2009; Kim et al., 2011), managerial career concerns (Andreou et al., 2017), as well as the broader information environment (An & Zhang, 2013; Callen & Fang, 2015b; Kim et al., 2019; Xu et al., 2019).

The rest of the paper is presented as follows. The theoretical foundation and hypothesis formulation are outlined in Section 2. The data and techniques used to estimate the impact of DBN legislation on crash risk and bad news hoarding are described in Section 3. Section 4 presents our empirical analyses, whereas Section 5 summarizes and presents avenues for future research.

2 | BACKGROUND LITERATURE AND HYPOTHESIS DEVELOPMENT

There is well-established literature examining the impact that mandatory financial disclosures have on a firm's disclosure regime. Much of it has been framed from the perspective of whether these disclosures serve as a complement to (Ball et al., 2012; Li & Yang, 2016), or a substitute for (Friedman et al., 2022; Heinle et al., 2022; Noh et al., 2019), voluntary disclosures. In either case, the research tends to support the view that mandatory financial disclosures generally improve the firm information environment to the benefit of investors (Greenstone et al., 2006; Li & Yang, 2016). There are, however, arguments that they can reduce price informativeness by changing the incentives for investors to collect information (Bird et al., 2021).

Separate to the studies examining the impact of financial disclosure regimes on firm and investor behavior, less is known about the effects of mandatory nonfinancial disclosures on firms (Leuz & Wysocki, 2016). One exception is Christensen et al. (2017), who showed that mandatory social responsibility disclosures for SEC-registered mine owners reduce mining injuries and labor productivity. Another example is Chen et al. (2018), who found that the introduction of mandatory reporting of corporate social responsibility activities by firms in China reduces their emissions and decreases overall firm profitability.

To expand our understanding of the impact that mandatory nonfinancial disclosure rules can have on firms, we examine the impact that state-based consumer protection laws (relating to data breaches) have on crash risk and managerial incentives to withhold negative information. These laws require the reporting of, essentially, only negative news and so impose an asymmetric reporting requirement on firms. Until 2003, firms had no obligation to disclose data breaches, implying that they could manage to avoid the financial and reputational costs associated with a disclosure. Although it is possible to mitigate some of the risk a firm is exposed to by investing more in data security (Murciano-Goroff, 2019), the costs to eliminate it are high from both a resource and an effort perspective. Some of the direct costs can be offset through cyber insurance, but this compensation does not extend to the substantial indirect costs, including the potential negative effect on long-term shareholder net worth (Kamiya et al., 2021).

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

We argue that DBN laws will alter the managerial incentives to hoard other negative news. This is based on the fact that declaring data breaches comes at a large cost for the firm. The costs to announcing data breaches include a significant decline in firm stock price (Kamiya et al., 2021), an adverse impact for the cost of raising capital (Huang & Wang, 2021), as well as longer term effects for firms' reputation and customer bases (Ablon et al., 2016). The DBN laws generally require firms to report data breaches to the affected parties and a state authority, which is usually the Attorney-General's office. They also impose deadlines for disclosing the data breaches, which usually range from 30 to 45 days. Firms can, of course, choose to violate the law but in doing so they generally risk facing penalties and/or the Attorney-General's office enforcing the law.[5] In addition, the reputation risk of a firm not disclosing a data breach and being caught later is likely a substantial deterrent to not disclosing. For example, Amir et al. (2018) found that cyberattacks that are concealed have a much more significant impact on equity values in the month when they are discovered compared to the impact of disclosed attacks.

Given that managers must now account for the cost associated with a firm possibly needing to declare a data breach in the future, we argue that the incentive structure of managers to release other types of negative news will change. Prior research has already shown that managers become more risk averse and increase firm cash holdings when faced with unhedgeable risks (Froot & Stein, 1998). Indeed, Garg (2020) found that this is the case when firms experience increased cybersecurity risk, and Boasiako and Keefe (2021) noticed the same when mandatory DBN laws are introduced. Additionally, after data breaches, firms increase cybersecurity risk factor disclosures in their annual reports (Chen et al., 2023).

Despite managers' increased risk aversion and investment in information security, there is no assurance that a data breach will be completely prevented, and they may still face liability if one occurs. Given that the disclosure of a data breach incident will undoubtedly result in a notable negative investor sentiment toward the company, any negative news that follows could be exacerbated by this sentiment, as noted by Mian and Sankaraguruswamy (2012). Therefore, managers may choose to withhold unfavorable information to avoid market overreactions. Similarly, companies with a high probability of facing a cyberattack may face adverse sentiment, even without experiencing an actual breach. This negative sentiment could stem from the data breach incidents of their industry peers, particularly after the implementation of DBN laws. As a result, managers of those firms could be motivated to strategically manipulate market perception by concealing unfavorable news as well.

In line with this perspective, Chen et al. (2019) discovered that opportunistic insider selling tends to increase when companies are required to reveal a data breach. As a result, managers have a further reason to be shortsighted and withhold more negative news. Furthermore, Liu and Ni (2024) showed that companies tend to engage in more real earnings management after the introduction of mandatory DBN laws. Taken together, these arguments suggest that the motivation to conceal adverse information increases following the implementation of DBN laws, thereby elevating the risk of stock price crashes. Thus, our hypothesis is as follows:

H1. The adoption of mandatory DBN laws results in a heightened stock price crash risk for firms due to their accumulation of more negative information.

3 | DATA AND METHOD

3.1 | Sample selection

Our sample consists of firm-level data from Compustat and the Center for Research in Security Prices (CRSP) from January 2001 to December 2019. We follow Huang and Wang (2021) and use enactment dates of the DBN law in

[5] For further details of each state's DBN laws, both I.T. Governance (see www.itgovernanceusa.com) and Perkinscoie (see https://www.perkinscoie.com) provide informative summaries.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

each state to identify when a firm, whose headquarters are in that state, is affected by the law.[6] Appendix 1 reports the effective dates for each state where a DBN law is enacted that requires those affected by a data breach to be notified.[7] The first occurs approximately 2 years after the start of our sample period (1 July 2003 for California), and the last occurs one and a half years before the end of our sample period (1 July 2018 for South Dakota). We follow the prior literature (Boasiako & Keefe, 2021; Huang & Wang, 2021; Liu & Ni, 2024) and remove firms from the heavily regulated financial and utility industries as they must adhere to different disclosure practices.[8] We also exclude firms that relocate their headquarters or state of incorporation during the sample period. This ensures that our sample does not include firms that may have migrated to states to suit their preferred legal environment when the DBN laws were enacted. After dropping observations with missing data and winsorizing the data at the 1 and 99 percentiles to remove outliers, our final sample consists of 37,987 firm-year observations from 4074 unique firms.[9]

3.2 | Method

To test for crash risk, we follow the same methodological setup that Callen and Fang (2015a) used, whose variable of interest is also at the state-level. We follow them and use three of the most commonly applied stock price crash risk measures based on firm-specific returns estimated from the residuals of an expanded market model (Hutton et al., 2009). The model is given in the following equation and is estimated on a rolling fiscal year basis:

$$r_{i,t} = a_i + \beta_{1,i} r_{m,t-1} + \beta_{2,i} r_{j,t-1} + \beta_{3,i} r_{m,t} + \beta_{4,i} r_{j,t} + \beta_{5,i} r_{j,t+1} + \beta_{6,i} r_{m,t+1} + \varepsilon_{i,t} \tag{1}$$

where a_i is the intercept, $r_{i,t}$ is the daily stock return for firm i, $r_{m,t}$ is the return of the CRSP value-weighted market index on day t, $r_{j,t}$ is the return on the value-weighted industry index when applying two-digit standard industry classification codes (SIC), and $\varepsilon_{i,t}$ is the error term. To account for nonsynchronous trading, the model includes lag and lead market and industry returns (Dimson, 1979). Firm-specific returns ($D_{i,t}$), for firm i, are then generated using the residuals derived from Equation (1). We also log transform these regression residuals to address skewness as per the following equation (Chen et al., 2001):

$$D_{i,t} = \ln\left(1 + \varepsilon_{i,t}\right) \tag{2}$$

Using the above generated firm-specific returns, we calculate our first stock price crash risk measure, negative skewness (*NSKEW*), using the following formula:

$$NSKEW_i = -\left[n(n-1)^{\frac{3}{2}} \sum D_{i,t}^3\right] / \left[(n-1)(n-2)\left(\sum D_{i,t}^2\right)\right]^{\frac{3}{2}} \tag{3}$$

where n is the number of observations during the fiscal year t.

The second measure we compute is down-to-up volatility (*DUVOL*). For each fiscal year, we first split returns for each firm i into down-day (return is less than the annual mean for firm i) and up-day (return is greater than the annual mean for firm i) categories. We then follow Equation (4) to compute *DUVOL*, which is defined as the natural logarithm

[6] We determine the state of a firm's headquarters based on Edgar electronic filings and from Bill McDonald's website (https://sraf.nd.edu/data/).

[7] We track dates for relevant DBN law statutes and their amendments using the Lexis Advance database and information available from IT Governance (see www.itgovernanceusa.com) and Perkinscoie (see https://www.perkinscoie.com).

[8] We use the Fama and French 49 Industry classification schema, with financials (FF49 codes 45–48) and utilities (FF49 code 31) being dropped.

[9] A sample reconciliation table is provided in Table O1 of the Supporting Information Appendix.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of the ratio of the down-day group's standard deviation to the up-day group's standard deviation:

$$DUVOL_i = \log\left(\left[(n_u - 1)\sum_d D_{i,t}^2\right] / \left[(n_d - 1)\right]\sum_u D_{i,t}^2\right) \tag{4}$$

where n_u is the number of observations in the up-group, and n_d is the number of observations in the down-group.

The final metric we use, *COUNT*, is the difference in the number of times firm-specific returns are above, relative to those that are below, k standard deviations of the mean. We set k to generate frequencies of 0.1% in the log-normal distribution (Hutton et al., 2009; Kim et al., 2011).

To examine the impact that DBN laws have on crash risk, we employ a DiD model. Our treatment (control) group includes all firms that are (are not) headquartered in a state that has an active DBN law in year t. We then apply the following baseline regression model:

$$CRASH\ RISK_{i,t+1} = \beta_0 + \beta_1 DBNL_{i,t} + \delta X_{i,t} + Year_t + Firm_i + \varepsilon_{i,t} \tag{5}$$

where i and t represent firm and year, respectively. The dependent variable, $CRASH\ RISK_{i,t+1}$, represents one of the three crash risk measures ($NSKEW_{t+1}$, $DUVOL_{t+1}$ or $COUNT_{t+1}$) for firm i. The independent variable of interest is the indicator variable $DBNL_{i,t}$. It is assigned a value of one to all firms headquartered in a state that has enacted a DBN law in year t, and zero otherwise.

In Equation (5), $X_{i,t}$ is a vector of firm-level controls. To be consistent with the prior literature, we use a set of firm-level controls similar to Callen and Fang (2015a). This is also consistent with Chen et al. (2001). We control for prior period crash risk using $NSKEW_t$; $KURTOSIS_t$ to capture the kurtosis of firm-specific daily returns; RET_t to represent cumulative firm-specific daily returns in the fiscal year; $STDEV_t$ to measure the standard deviation of firm-specific daily returns in the fiscal year; $SIZE_t$, measured as the logarithm of the market value of equity; the market to book ratio (MTB_t); $LEVERAGE_t$, measured as total long-term debt divided by total assets; ROE_t, measured as income before extraordinary items divided by the book value of equity; share turnover ($DTURN_t$); and a measure of firm opacity ($OPACITY_t$) based on the 3-year moving sum of discretionary accruals (Hutton et al., 2009). We also include an indicator variable, LIT_RISK_t, to account for industry level litigation risk which, following Callen and Fang (2015a), is set to equal one for firms classified in the fiscal year as being in the biotechnology, computer, electronics or retail industries, and zero otherwise.

Additionally, we control for the number of analysts following a firm ($ANALYST_t$) as Chen et al. (2001) demonstrated that analysts improve the firm's information environment, thereby affecting crash risk. Moreover, for similar reasons, we control for the proportion of institutional ownership ($INST_OWN_t$) within the firm (An & Zhang, 2013). Finally, as Callen and Fang (2015a) showed that religious adherence can influence crash risk at the state level, we include the number of religious adherents as a percentage of the state's total population ($RELIGION_t$).[10] A detailed description of how all of the above variables are calculated is provided in Appendix 2.

In our baseline model, we also include firm fixed effects to account for omitted firm-specific factors and year fixed effects to control for unobservable time-variant effects. As our variable of interest, $DBNL_{i,t}$, varies by state, we use two-way clustering of the standard errors by headquarters state and by year to account for possible serial dependence in the cross section and in time.

[10] We construct the religious adherence variable using data from the Association of Religion data archives. See http://www.thearda.com/Archive/ChCounty.asp.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Summary statistics.

Panel A. Descriptive statistics.							
	Mean	Standard deviation	*Min*	*P25*	*Median*	*P75*	*Max*
$DBNL_t$	0.6008	0.4897	0.0000	0.0000	1.0000	1.0000	1.0000
Dependent variables							
$NSKEW_{t+1}$	−0.0016	1.6567	−14.1282	−0.6966	−0.1461	0.4621	13.9611
$DUVOL_{t+1}$	−0.0506	0.3371	−2.1478	−0.2555	−0.0697	0.1304	1.9312
$COUNT_{t+1}$	−0.1308	1.3444	−5.0000	−1.0000	0.0000	1.0000	6.0000
Control variables							
$KURTOSIS_t$	9.1193	12.1600	0.3344	2.3222	4.6612	10.3657	71.7922
RET_t	−0.2201	0.2070	−0.9163	−0.3038	−0.1453	−0.0684	−0.0110
$STDEV_t$	0.0299	0.0182	0.0067	0.0168	0.0250	0.0380	0.0977
MTB_t	2.8514	4.5699	−14.1330	1.1196	1.9379	3.4177	29.5176
$LEVERAGE_t$	0.1746	0.1960	0.0000	0.0004	0.1190	0.2836	0.8914
ROE_t	−0.0477	0.7409	−4.7338	−0.0437	0.0690	0.1377	2.7069
$SIZE_t$	14.6222	2.6953	7.6962	12.8063	14.9289	16.5743	19.9098
$DTURN_t$	0.0145	1.0665	−4.1088	−0.3206	−0.0030	0.3185	4.5847
$OPACITY_t$	0.1829	0.1690	0.0153	0.0786	0.1324	0.2232	1.0275
$RELIGION_t$	0.5037	0.0767	0.3144	0.4481	0.5194	0.5632	0.6822
LIT_RISK_t	0.3278	0.4694	0.0000	0.0000	0.0000	1.0000	1.0000
$INST_OWN_t$	0.5784	0.3081	0.0005	0.3154	0.6460	0.8453	1.0000
$ANALYST_t$	1.8063	1.0482	0.0000	1.0986	1.9459	2.6391	3.6636

Panel B. Univariate tests					
	Pre DBNL	Post DBNL	Difference	*t*-Stat	*p*-Value
$NSKEW_t$	−0.0197	0.0105	0.0302	1.7408	0.0817
$DUVOL_t$	−0.0600	−0.0443	0.0156	4.4280	0.0000
$COUNT_t$	−0.1808	−0.0976	0.0832	5.9099	0.0000

Note: Panel A reports descriptive statistics for the variables used in the baseline regressions of this study. The sample consists of 37,987 firm-year observations from 2001 to 2019. All variables are defined in Appendix 2. Panel B compares the average crash risk measures ($NSKEW_t$, $DUVOL_t$ and $COUNT_t$) for firms prior to, and after the state, the firm is headquartered in enacts DBN laws. The *t*-statistics and *p*-values are from difference in means tests.

4 | EMPIRICAL RESULTS

4.1 | Descriptive statistics and univariate results

Panel A of Table 1 presents the descriptive statistics for the variables used in our baseline crash risk regressions. The mean of the variable of interest, $DBNL_t$, indicates that 60.1% of the sample is from observations in states where DBN laws have been enacted. Panel B provides univariate analysis of our crash risk proxies before and after a state adopts DBN laws. We calculate the average of $NCSKEW_t$, $DUVOL_t$, $COUNT_t$ for both the pre- and post-DBN law adoption period and report these figures along with the difference in the values before and after the event. The fourth column displays the *t*-statistics for the differences in means, showing a significant rise in the crash risk measures following a state's adoption of DBN laws, significant at the 1% level for $DUVOL_t$ and $COUNT_t$ (increases of 0.016 and 0.083,

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

respectively). $NCSKEW_t$ also rises by 0.030, which is significant at the 10% level. These results provide model-free support for our hypothesis.[11]

4.2 | Main results

Table 2 reports our baseline results from regressing Equation (5). Our three crash risk measures are regressed against our variable of interest, $DBNL_t$, with fixed effects in the first three models. Models 4–6 then include our full set of control variables. In support of our hypothesis, the coefficients of $DBNL_t$ are positive and significant to at least the 1% level of significance when the dependent variable is either $DUVOL_{t+1}$ or $COUNT_{t+1}$, and significant at the 5% level for $NSKEW_{t+1}$. We follow the approach of Xu et al. (2021) to infer economic significance and analyze the proportional increase in crash risk that occurs when a state adopts a DBN law relative to the dispersion of each crash risk measure. Using the coefficient results from Models 4–6, crash risk increases by 5.64%, 7.36% and 6.83% of the standard deviation of the respective $NSKEW_{t+1}$, $DUVOL_{t+1}$ and $COUNT_{t+1}$ measures.[12]

We next show results from trend analysis of our variable of interest, $DBNL_t$. If $DBNL_t$ is capturing a significant change in the future crash risk of a firm as a direct result of a state adopting a DBN law, any lagged DBN law indicator should be insignificant. If this is not the case, the validity of our baseline findings is questionable, as there would be evidence of a pretreatment trend in crash risk prior to the DBN laws being adopted. This would make the DiD parallel trends assumption invalid. To test for evidence of a pre-trend, we replace $DBNL_t$ in Equation (5) with several indicator variables. $DBNL_0$ is equal to one during the year of a state's DBN law adoption, and otherwise zero. $DBNL_{t-1}$ and $DBNL_{t-2}$ are equal to one for the year and 2 years prior to adoption, respectively. Otherwise, the values are zero. We also include two forward-looking DBNL variables. $DBNL_{t+1}$ is equal to one the year after DBN law adoption, whereas $DBNL_{\geq t+2}$ is equal to one for 2 years succeeding the adoption and thereafter. Otherwise, the values for these indicator variables are set to zero. Including the lead variables will provide information on the post-adoption dynamics of DBN laws.

Table 3 presents the results and shows that the coefficients for the lagged variables are all statistically insignificant, implying that there is no evidence of a pretreatment trend in crash risk before the adoption of DBN laws. Rather, the impact of a state adopting a DBN law on a firm's crash risk is felt the year after it is adopted, as all the coefficient estimates of $DBNL_0$ are significant in the regressions at the 1% level. Furthermore, based on the significance of the coefficients of the forward-looking DBN law indicators, the consequence of implementing DBN laws appears to have an impact on crash risk in future years.

As a means to test that our baseline results are specifically being driven by the adoption of DBN laws in the states that firms are headquartered, we conduct permutation inference tests. We first randomly reassign a firm's location to another state. We then reestimate our baseline regressions. If our results are driven by unobserved shocks that coincide with a state adopting DBN laws, we should still find the coefficient of $DBNL_t$ to be significant even when firms are assigned to another state. To strengthen the test, we repeat the experiment 1000 times and record the coefficient and t-statistic values of $DBNL_t$ from each iteration.

Figure 1 shows three histograms of the coefficients for $DBNL_t$ derived from the tests performed for each of the three crash risk measures. The coefficients are centered close to the value of zero. Two vertical lines are included in each histogram to represent the estimated coefficient obtained in our baseline results (i.e., Models 4–6 in Table 2) and the estimated coefficient multiplied by −1. In reviewing the t-statistics, only 47 out of 1000 placebo t-values are larger than the true t-statistic for $NSKEW_{t+1}$, with 8 for $DUVOL_{t+1}$, and 16 for $COUNT_{t+1}$. Taken together, these analyses

[11] Table O2 of the online appendix presents a Pearson correlation matrix of all the variables used in the crash risk regressions. Supporting the univariate results, $DBNL_t$ is positively correlated with all three crash risk measures, with $DUVOL_t$ and $COUNT_t$ being significant at the 1% level.

[12] These figures are based on dividing the $DBNL_t$ coefficients in each regression by the standard deviation of the relevant crash risk measure reported in Table 1. Following Xu et al. (2021), we choose this method instead of basing economic significance on the means of the variables as they are all close to the value of zero, implying that even minor numerical changes in these values will lead to large percentage changes and likely overestimate the size effect.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 Baseline regression results.

	$NSKEW_{t+1}$ (1)	$DUVOL_{t+1}$ (2)	$COUNT_{t+1}$ (3)	$NSKEW_{t+1}$ (4)	$DUVOL_{t+1}$ (5)	$COUNT_{t+1}$ (6)
$DBNL_t$	0.0956*	0.0259***	0.0959***	0.0935**	0.0248***	0.0918***
	(0.05)	(0.01)	(0.03)	(0.04)	(0.01)	(0.03)
$NSKEW_t$				−0.1069***	−0.0211***	−0.0471***
				(0.01)	(0.00)	(0.01)
$KURTOSIS_t$				0.0022**	0.0007***	0.0022***
				(0.00)	(0.00)	(0.00)
RET_t				−0.2959	0.0149	0.2104
				(0.55)	(0.10)	(0.22)
$STDEV_t$				−15.3559**	−2.8764**	−5.9640*
				(6.89)	(1.28)	(2.91)
MTB_t				0.0058**	0.0016***	0.0061**
				(0.00)	(0.00)	(0.00)
$LEVERAGE_t$				0.0393	−0.0067	−0.1313**
				(0.09)	(0.02)	(0.05)
ROE_t				0.0404**	0.0101***	0.0384**
				(0.02)	(0.00)	(0.01)
$SIZE_t$				0.1553***	0.0374***	0.1150***
				(0.01)	(0.00)	(0.01)
$DTURN_t$				0.0366***	0.0081***	0.0195*
				(0.01)	(0.00)	(0.01)
$OPACITY_t$				0.1028	0.0127	0.0875
				(0.10)	(0.02)	(0.07)
$RELIGION_t$				0.3116	0.0400	−0.1619
				(0.66)	(0.13)	(0.91)
LIT_RISK_t				−0.0339	−0.0308	−0.2288
				(0.24)	(0.03)	(0.25)
$INST_OWN_t$				0.1592**	0.0371**	0.0508
				(0.07)	(0.02)	(0.07)
$ANALYST_t$				0.1319***	0.0307***	0.1088***
				(0.02)	(0.01)	(0.02)
Intercept	−0.0590**	−0.0661***	−0.1884***	−2.4719***	−0.6214***	−1.7417***
	(0.03)	(0.01)	(0.02)	(0.38)	(0.07)	(0.41)
Firm F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	37,987	37,987	37,987	37,987	37,987	37,987
Adjusted R^2	0.0329	0.0495	0.0424	0.0691	0.0948	0.0656

Notes: This table reports panel OLS regression results showing the impact that DBN laws have on crash risk based on Equation (5). The sample consists of firm-year observations for the period 2001–2019. The first three columns are fixed effects models, whereas the remaining three columns also include control variables. All variables are defined in Appendix 2. Standard errors are in parentheses and are two-way clustered by headquarter state and by year.
***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 Trend analysis.

	$NSKEW_{t+1}$ (1)	$DUVOL_{t+1}$ (2)	$COUNT_{t+1}$ (3)
$DBNL_{t-2}$	0.0539	0.0160	0.0432
	(0.04)	(0.01)	(0.03)
$DBNL_{t-1}$	0.0906	0.0146	0.0575
	(0.06)	(0.01)	(0.05)
$DBNL_0$	0.2015***	0.0467***	0.1574***
	(0.05)	(0.01)	(0.03)
$DBNL_{t+1}$	0.0936**	0.0219***	0.0827*
	(0.04)	(0.01)	(0.05)
$DBNL_{t\geq 2}$	0.1485*	0.0418***	0.1709***
	(0.07)	(0.01)	(0.05)
$NSKEW_t$	−0.1069***	−0.0211***	−0.0471***
	(0.01)	(0.00)	(0.01)
$KURTOSIS_t$	0.0022**	0.0007***	0.0022***
	(0.00)	(0.00)	(0.00)
RET_t	−0.2852	0.0168	0.2162
	(0.55)	(0.10)	(0.22)
$STDEV_t$	−15.3123**	−2.8669**	−5.9245*
	(6.91)	(1.28)	(2.93)
MTB_t	0.0058**	0.0016***	0.0061**
	(0.00)	(0.00)	(0.00)
$LEVERAGE_t$	0.0408	−0.0062	−0.1292**
	(0.09)	(0.02)	(0.05)
ROE_t	0.0406**	0.0102**	0.0388**
	(0.02)	(0.00)	(0.01)
$SIZE_t$	0.1546***	0.0372***	0.1147***
	(0.01)	(0.00)	(0.01)
$DTURN_t$	0.0372***	0.0082***	0.0199**
	(0.01)	(0.00)	(0.01)
$OPACITY_t$	0.1034	0.0125	0.0866
	(0.10)	(0.02)	(0.06)
$RELIGION_t$	0.3490	0.0475	−0.1397
	(0.65)	(0.13)	(0.90)
LIT_RISK_t	−0.0337	−0.0313	−0.2316
	(0.25)	(0.03)	(0.25)
$INST_OWN_t$	0.1607**	0.0375**	0.0515
	(0.07)	(0.02)	(0.07)
$ANALYST_t$	0.1315***	0.0306***	0.1082***
	(0.02)	(0.01)	(0.02)

(Continues)

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 (Continued)

	$NSKEW_{t+1}$	$DUVOL_{t+1}$	$COUNT_{t+1}$
	(1)	(2)	(3)
Intercept	−2.5052***	−0.6295***	−1.7768***
	(0.36)	(0.07)	(0.40)
Firm F.E.	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes
Observations	37,987	37,987	37,987
Adjusted R^2	0.0693	0.0950	0.0658

Notes: This table reports trend analysis based on replacing $DBNL_t$ in Equation (5) with a set of indicator variables to capture pre- and post-DBN law enactment effects. $DBNL_0$, $DBNL_{t-1}$ and $DBNL_{t-2}$ are equal to the value of one during the year of a state's enactment of a DBN law, the year prior to adoption and 2 years prior to adoption, respectively, and otherwise zero. $DBNL_{t+1}$ is equal to one the year following DBN law adoption, whereas $DBNL_{\geq t+2}$ is equal to the value of one from 2 years and onward following adoption, and zero otherwise. Standard errors are in parentheses and are two-way clustered by headquarter state and by year. All variables are defined in Appendix 2.
***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

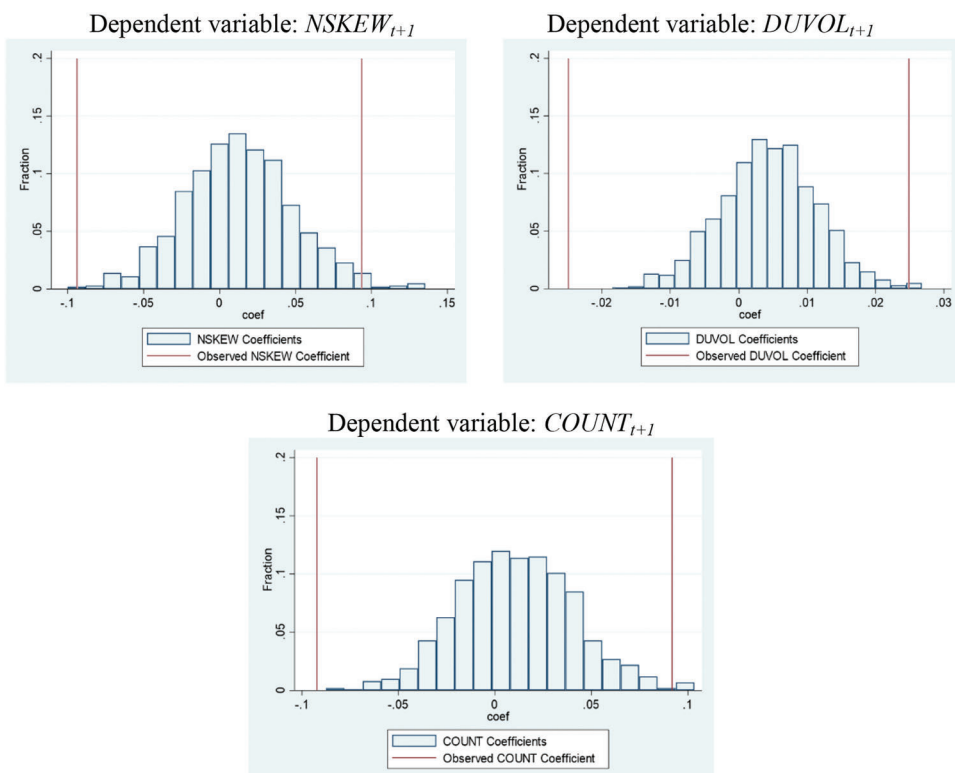

FIGURE 1 Histogram of coefficients from permutation inference tests. The histogram on the top left plots the distribution of the $DBNL_t$ coefficients generated from running 1000 iterations of our baseline regression (Equation 5) where $NCSKEW_{t+1}$ is the dependent variable. For each iteration, firms are randomly assigned to a state other than the one they are headquartered in. The vertical lines represent the coefficients generated from using the actual, non-randomly assigned, data, as well as the estimated coefficient multiplied by −1. The histogram on the top right shows the results where the dependent variable is $DUVOL_{t+1}$. The bottom histogram shows the results where the dependent variable is $COUNT_{t+1}$.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

indicate that our baseline results are specifically related to the dates that states adopt DBN laws and how that relates to where the firms are located.[13]

4.3 | Channel analysis

We now turn our attention to investigate the underlying mechanism that leads mandatory DBN laws to increase crash risk. In particular, we examine whether firms are hoarding more bad news following the adoption of DBN laws.

We approach testing for the hoarding of bad news from several different angles. We first examine the change in residual short interest as a proxy for managerial private bad news hoarding. As high short interest is a good predictor of negative information that is not already reflected in stock prices (Christophe et al., 2004; Desai et al., 2002) and thereby also leads to increased crash risk (Callen & Fang, 2015b), Bao et al. (2019) constructed a residual short interest measure that accounts for short selling factors unrelated to bad news. We replicate their methodology and capture the residuals ($\varepsilon_{i,t}$) from the following regression:

$$SI_{i,t} = \alpha + \beta_1 INST_OWN_{i,t} + \beta_2 CONVERT_{i,t} + \beta_3 TREND_t + \varepsilon_{i,t} \tag{6}$$

where SI_t is short interest of a firm at the end of fiscal year t, $INST_OWN_t$ is institutional ownership within the firm, $CONVERT_t$ is an indicator variable taking the value of one if the firm has convertible bonds or convertible preferred stocks outstanding, and zero otherwise, and $TREND_t$ is a time-trend variable. These variables are included in the regression to account for the impact that the supply of loanable shares, hedge-based shorting and a general upward time trend in shorting have on short interest, respectively (Bao et al., 2019).

If DBN laws have influenced the amount of bad news managers withhold from the market, we should find a significant increase in the residual short interest following the enactment of DBN laws. The first column of Table 4 reports the results of regressing residual short interest ($ResSI_t$) on $DBNL_t$ while using a set of firm-specific controls used in the prior literature that examines disclosure changes following a law change (see Bourveau et al., 2018). The coefficient of $DBNL_t$ ($\beta = 0.0023$) is significant at the 5% level, indicating that following the enactment of DBN laws, bad news hoarding, as proxied using the residual short interest measure, increases by 5.2% of its standard deviation.[14]

Another possible proxy for bad news hoarding is the incidence of securities class action lawsuits firms face. If managers are not disclosing price-sensitive information in a timely manner, it may expose the firm to legal action. We collect data from the Stanford Law School Securities Class Action Clearinghouse filings database and construct an indicator variable, SCA_t, equal to one if the firm faces a securities class action in the fiscal year, and otherwise zero. Using this measure as our dependent variable, we reestimate the model, and the results, reported in column (2) of Table 4, suggest that there is an increased risk of litigation following the adoption of the laws, albeit only at the 10% level of significance.

Turning our attention to corporate disclosure, we test for any potential change in firms' management guidance and voluntary 8K filings following the adoption of DBN laws. Management forecasts are a principal way for managers to disclose private, future-oriented financial information to capital markets (Healy & Palepu, 2001) and carry a significant component of a firm's overall information environment (Hirst et al., 2007). Furthermore, management forecasts are a form of financial disclosure with the highest level of information content (Beyer et al., 2010). If DBN laws discourage managers from divulging negative news, we should find a decline in pessimistic earnings guidance and possibly a rise in optimistic forecasts.

[13] In Table O3 of the Supporting Information Appendix, we check to see if our results are not merely an artifact of increased crash risk coming from firms that have experienced a data breach by excluding all firms that are recorded to have had a data breach. We also show the regression results for when we exclude firms in the same Fama French 49-industry classification for the year a firm experiences a breach to ensure that our results are not driven solely by peer effects.

[14] The mean and standard deviation of $ResSI_t$ are −0.0007 and 0.0442, respectively.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 Residual short interest and incidence of class action lawsuits.

	$ResSI_{t+1}$ (1)	SCA_{t+1} (2)
$DBNL_t$	0.0023**	0.0039*
	(0.00)	(0.00)
$INST_OWN_t$	0.0128***	−0.0044
	(0.00)	(0.01)
$SIZE_t$	0.0058***	0.0045***
	(0.00)	(0.00)
MTB_t	−0.0001	0.0005*
	(0.00)	(0.00)
ROE_t	0.0007**	0.0025*
	(0.00)	(0.00)
$LITIG_RISK_t$	−0.0027	−0.0452*
	(0.01)	(0.02)
RET_t	−0.0189***	0.0046
	(0.00)	(0.01)
$LOSS_t$	0.0029***	0.0007
	(0.00)	(0.00)
$EARNINGS_VOL_t$	0.0075***	−0.0121*
	(0.00)	(0.01)
Intercept	−0.0994***	−0.0354**
	(0.01)	(0.01)
Firm F.E.	Yes	Yes
Year F.E.	Yes	Yes
Observations	36,644	36,693
Adjusted R^2	0.4784	0.0137

Notes: This table reports results from examining the impact that DBN laws have on firms' residual short interest (*ResSI*), and on securities class action litigation (*SCA*). Standard errors are in parentheses and are two-way clustered by headquarters state and by year. All variables are defined in Appendix 2.
***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

To test the above expectations, we count the number of times earnings guidance in the fiscal year is either above or below actual earnings to compute two management guidance variables: $Optimistic_t$ and $Pessimistic_t$, respectively.[15] Columns 1–3 of Table 5 report the results where they are used as dependent variables along with a third measure, $O\text{-}P_Diff_t$, which is the difference between the two. After the enactment of DBN laws, management forecasts become more optimistic (Column 1) and less pessimistic (Column 2), with the amount of positive relative to negative management forecasts (Column 3) significantly increasing at the 5% level. The $DBNL_t$ coefficient in the $O\text{-}P_Diff_{t+1}$ regression ($\beta = 0.0981$) indicates that after DBN law adoption, firms' earnings guidance tends, on average, to be 7.7% relatively more positive.[16]

[15] We use the mid-point if a forecast range is provided instead of a point estimate.

[16] The mean and standard deviation of *O-P_Diff* are −0.4943 and 1.2751, respectively.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 5 Management guidance and voluntary 8K filings.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	$Optimistic_{t+1}$	$Pessimistic_{t+1}$	$O\text{-}P_Diff_{t+1}$	$POS8K_{t+1}$	$NEG8K_{t+1}$	$POS8K\text{–}NEG8K_{t+1}$	$VOL8K_{t+1}$
$DBNL_t$	0.0444*	−0.0536**	0.0981**	0.0088**	0.0127	−0.0039	0.0155**
	(0.03)	(0.02)	(0.04)	(0.00)	(0.01)	(0.01)	(0.01)
$INST_OWN_t$	0.0747	0.0402	0.0345	0.0331	0.0423	−0.0091	0.0549
	(0.06)	(0.06)	(0.10)	(0.03)	(0.03)	(0.02)	(0.04)
$SIZE_t$	0.0769***	−0.0219***	0.0988***	−0.0038	−0.0012	−0.0025	−0.0026
	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)	(0.00)	(0.01)
MTB_t	−0.0034*	0.0035*	−0.0069**	−0.0013	−0.0007	−0.0005	−0.0018
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
ROE_t	0.0298**	−0.0019	0.0318	−0.0076**	−0.0063	−0.0013	−0.0112**
	(0.01)	(0.01)	(0.02)	(0.00)	(0.00)	(0.00)	(0.01)
RET_t	0.4572***	0.2276***	0.2296*	0.0256	−0.0372	0.0628*	−0.0222
	(0.08)	(0.08)	(0.13)	(0.04)	(0.05)	(0.03)	(0.06)
LIT_RISK_t	0.6788*	0.3588	0.3200	0.0261	0.2401	−0.2140*	0.2466
	(0.40)	(0.44)	(0.68)	(0.11)	(0.16)	(0.10)	(0.21)
$LOSS_t$	−0.0639***	−0.0332	−0.0306	0.0162	0.0146	0.0015	0.0229
	(0.02)	(0.02)	(0.04)	(0.01)	(0.01)	(0.00)	(0.01)
$EARNINGS_VOL_t$	0.0785	−0.0820	0.1605	−0.0746**	−0.1016***	0.0269	−0.1296***
	(0.09)	(0.09)	(0.16)	(0.03)	(0.03)	(0.02)	(0.04)
Intercept	−0.7139***	1.5249***	−2.2388***	0.4023***	0.2779***	0.1244**	0.5064***
	(0.21)	(0.21)	(0.35)	(0.08)	(0.08)	(0.05)	(0.11)
Firm F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,673	13,673	13,673	35,407	35,407	35,407	35,407
Adjusted R^2	0.1913	0.3336	0.1917	0.4529	0.4494	0.0107	0.5123

Notes: This table reports results from examining the impact that DBN laws have on management guidance and voluntary 8K filings. Standard errors are in parentheses and are two-way clustered by headquarters state and by year. All variables are defined in Appendix 2.
***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

As an alternative to examining earnings guidance, we also analyze changes in voluntary 8K filings following DBN law adoption to assess shifts in disclosure patterns. We first calculate a count measure that is one plus the logarithm of the number of voluntary 8K filings in a fiscal year ($VOL8K_t$) to proxy for management's willingness to update investors. We then classify each 8K as imparting good or bad news by calculating the 3-day ($t-1$ to $t+1$) CAR window around filing dates based on a 120-day market model. Negative CARs indicate bad news filings ($NEG8K_t$), whereas positive CARs reflect good news ($POS8K_t$), with the difference being $POS8K\text{–}NEG8K_t$. Results in Table 5 show that DBN laws are associated with a significant increase in total voluntary 8K filings ($VOL8K_t$), principally driven by a significant increase in the disclosure of good news filings ($POS8K_t$). Together with our earlier earnings guidance findings, these results suggest that firms release more good news over bad news in voluntary disclosures post-DBN law adoption.

Recognizing that peer firms to those that have recently experienced a data breach are likely to be most affected in terms of the perceived risk of also being targeted, Panel A of Table 6 shows the impact on $Optimistic_t$, $Pessimistic_t$

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 6 Change in management guidance disclosure policy before and after data breaches.

Panel A. All peer firms					
	Peer group: FF49				
	Before	After	*Difference*	*t*-Stat	*p*-Value
$Optimistic_t$	0.0361	0.0261	0.0100	1.8202	0.0688
$Pessimistic_t$	0.0790	0.0354	0.0436	5.3682	0.0000
$O\text{-}P_Diff_t$	−0.0429	−0.0093	−0.0336	−3.4539	0.0006
	Peer group: FIC-25				
	Before	After	*Difference*	*t*-Stat	*p*-Value
$Optimistic_t$	0.0377	0.0279	0.0098	1.7231	0.0849
$Pessimistic_t$	0.0707	0.0418	0.0289	3.8005	0.0001
$O\text{-}P_Diff_t$	−0.0331	−0.0139	−0.0191	−2.0295	0.0424
Panel B. Non-DBNL peer firms					
	Peer group: FF49				
	Before	After	*Difference*	*t*-Stat	*p*-Value
$Optimistic_t$	0.0350	0.0357	−0.0007	−0.0720	0.9426
$Pessimistic_t$	0.0746	0.0246	0.0500	3.6349	0.0003
$O\text{-}P_Diff_t$	−0.0396	0.0111	−0.0507	−3.0120	0.0026
	Peer group: FIC-25				
	Before	After	*Difference*	*t*-Stat	*p*-Value
$Optimistic_t$	0.0345	0.0389	−0.0044	−0.4348	0.6637
$Pessimistic_t$	0.0651	0.0490	0.0161	1.2677	0.2050
$O\text{-}P_Diff_t$	−0.0306	−0.0102	−0.0205	−1.2576	0.2086
Panel C. DBNL peer firms					
	Peer group: FF49				
	Before	After	*Difference*	*t*-Stat	*p*-Value
$Optimistic_t$	0.0391	0.0236	0.0155	1.7848	0.0743
$Pessimistic_t$	0.0907	0.0382	0.0525	3.9840	0.0001
$O\text{-}P_Diff_t$	−0.0516	−0.0146	−0.0370	−2.3841	0.0172
	Peer group: FIC-25				
	Before	After	*Difference*	*t*-Stat	*p*-Value
$Optimistic_t$	0.0490	0.0247	0.0243	2.6779	0.0074
$Pessimistic_t$	0.0907	0.0397	0.0510	3.8772	0.0001
$O\text{-}P_Diff_t$	−0.0417	−0.0151	−0.0267	−1.6967	0.0898

Notes: This table reports results from examining the impact that data breaches have on the number of occurrences of optimistic management forecasts ($Optimistic_t$), pessimistic management forecasts ($Pessimistic_t$) and the difference between the two ($O\text{-}P_Diff_t$) on peer firms 3 years before and after the event occurs. FF49 represents firms that are considered peers based on belonging to the same Fama French 49 industry classification schema, whereas FIC-25 represents Hoberg and Phillips (2010, 2016) text-based classifications of firms that produce similar products. All variables are defined in Appendix 2.
***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and *O-P_Diff*$_t$ 3 years before and after a peer firm experiences a data breach. For robustness, we define peer firms in two distinct ways—first based on the Fama French 49 industry classification (FF49), and then based on Hoberg and Phillips' (2010, 2016) text-based classifications of firms that produce similar products. The FIC-25 peer group focuses on firms that are in a similar product market as the focal firm that has experienced a data breach. Focusing on *O-P_Diff*$_t$, the results indicate a significant increase in the amount of optimistic, relative to pessimistic, management forecasts released to the market, providing evidence that data breaches discourage the release of negative news.[17] In Panels B and C of Table 6, we reassess the impact of data breaches by segregating peer firms based on whether they operate in a state with DBN laws or not. Although the aforementioned results suggest that firms are more inclined to withhold unfavorable news following a peer's data breach announcement, those in DBN law states may be even more inclined to do so. The rationale being these firms would legally be obliged to disclose a subsequent data breach. The results reveal that *O-P_Diff*$_t$ becomes more positive irrespective of the firm location when FF49 peer firms are examined. Yet, when peer firms are defined as those in the same product market, the *O-P_Diff*$_t$ value only becomes more positive for firms operating in states with DBN laws.

4.4 | Cross-sectional analyses: managerial incentive and opportunity to hoard bad news

Our argument for why firms stockpile negative news following the enactment of DBN laws is based on managers having an incentive to do so. If this is the case, we should find cross-sectional variation in our sample as some firms will be more exposed to the impact of DBN laws than others, altering how incentivized managers will be to hide bad news. To test for this cross-sectional variation, we consider three factors that would alter the incentive structure for managers to hoard bad news following the enactment of the laws.

The first factor we examine relates to an important difference in the DBN laws enacted across the states. Although all DBN laws require a similar minimum level of disclosure and have mechanisms to ensure compliance, whether via the State Attorney-General's office or through the threat of civil penalties, these laws do not universally allow the affected party to take direct action themselves. Only 25 states provide a private cause of action within their DBN statutes, allowing for individuals to seek compensation through the courts in the event of a violation or simply to recover costs from those responsible. As Winn (2009) noted, doing so increases litigation risk through private and class action lawsuits. Therefore, firm litigation risk, as specifically related to the DBN laws, will be substantially higher where the DBN laws allow for a private cause of action. We expect that this increased litigation risk will further amplify the impact of DBN laws on crash risk, as managerial incentives to hoard bad news increase in line with the increasing costs of disclosing a data breach.

Using Lexis Advance and information provided by the National Conference of State Legislatures (NCSL), we trace when states introduce a private cause of action into their DBN laws.[18] Appendix 1 flags the states whose DBN laws allow for a private cause of action. We then reestimate our baseline model when splitting our firm-year observations into those exposed to a private cause of action and those that are not. The results are reported in Panel A of Table 7. We find that the *DBNL*$_t$ coefficient is statistically significant only for the subsample of firms exposed to DBN laws that allow a private cause of action.

Next, we utilize two measures to account for a firm's *ex-ante* risk of facing a data breach. If a firm is less likely to be exposed to a data breach, the need for managers to be concerned with the enactment of DBN laws will be reduced and their incentive to hoard bad news will be reduced. Our first measure to capture a firm's *ex-ante* data breach risk is industry based. Some industries, such as technology-based firms, have historically been more exposed to data

[17] It should be noted that including or excluding the firm that has experienced the data breach in the peer firm analysis does not materially change any of the results.

[18] Further information on the NCSL can be found at www.ncsl.org.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 7 Cross-sectional analysis: managers' incentive to hoard bad news.

Panel A. Private course of action						
	$NSKEW_{t+1}$		$DUVOL_{t+1}$		$COUNT_{t+1}$	
	(1)	(2)	(3)	(4)	(5)	(6)
	No	Yes	No	Yes	No	Yes
$DBNL_t$	0.0578	0.5228***	0.0165	0.0884***	0.0519	0.0195
	(0.06)	(0.13)	(0.01)	(0.02)	(0.06)	(0.11)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,633	16,078	21,633	16,078	21,633	16,078
Adjusted R^2	0.0789	0.0713	0.1155	0.0897	0.0780	0.0631
Panel B. Industry-based *ex-ante* breach risk						
	$NSKEW_{t+1}$		$DUVOL_{t+1}$		$COUNT_{t+1}$	
	(1)	(2)	(3)	(4)	(5)	(6)
	Low risk	High risk	Low risk	High risk	Low	High risk
$DBNL_t$	−0.0428	0.1467***	−0.0040	0.0369***	0.0153	0.1219**
	(0.06)	(0.04)	(0.01)	(0.01)	(0.04)	(0.05)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	11,737	26,217	11,737	26,217	11,737	26,217
Adjusted R^2	0.0773	0.0661	0.1044	0.0912	0.0736	0.0621
Panel C. TMT-based *ex-ante* breach risk (I.T. specialist in the top management team)						
	$NSKEW_{t+1}$		$DUVOL_{t+1}$		$COUNT_{t+1}$	
	(1)	(2)	(3)	(4)	(5)	(6)
	No	Yes	No	Yes	No	Yes
$DBNL_t$	0.0968*	−0.0346	0.0254***	0.0155	0.0979***	0.0374
	(0.05)	(0.19)	(0.01)	(0.04)	(0.03)	(0.21)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	31,371	1806	31,371	1806	31,371	1806
Adjusted R^2	0.0668	0.0856	0.0873	0.1160	0.0607	0.0641

Notes: This table reports regression results after subsampling the data. In Panel A, "Low Risk" firms are those located in states whose DBN laws do not allow for a private cause of action. Otherwise, they are placed in the "High Risk" category. In Panel B, firms are placed in the "Low Risk" category if in the preceding 3 years they are in an industry with data breach instances that are below the mean. Otherwise, they are placed in the "High Risk" category. In Panel C, firms are placed in the "Yes" category if, during the fiscal year, they have an information technology officer as part of their top management team. All variables are defined in Appendix 2. Standard errors are in parentheses and are two-way clustered by headquarters state and by year. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

breaches, particularly cyber breaches, than others. To account for this, we first collect all data breaches reported by Rosati and Lynn (2021) and the Privacy Rights Clearinghouse (PRC). For each year, we then calculate the 3-year historical average of data breaches by industry (2-digit SIC) and classify those industries that are above the average as having high *ex-ante* risk.[19] Panel B of Table 7 reports that the coefficients of $DBNL_t$ are only significant in the subsamples where industry *ex-ante* data breach risk is high.

In Panel C of Table 7, we consider the impact of having an information technology (I.T.) specialist as part of the top management team (TMT) on firm crash risk when DBN laws are adopted. As a proxy for firms that take information security more seriously, we divide our sample into those with an I.T. specialist on the TMT and those that do not have one. We use Execucomp and BoardEx to search for all TMTs of firms that have a Chief Information Security (or Technology) Officer.[20] If I.T. specialists on the TMT signal firms have invested in data security or, at least, treat it seriously in the organization, their firm's *ex-ante* risk should be reduced. The results support this notion as the coefficient of $DBNL_t$ is only significant in the subsample where there is no I.T. specialist on the TMT.

We next investigate the effect of the firm's information environment on moderating the impact that DBN laws have on crash risk. As the opportunity for managers to hide bad news will be reduced, if the information environment surrounding the firm is less opaque (Hong et al., 2017; Jin & Myers, 2006), we should find that this dampens the impact of DBN laws on crash risk for firms with lesser agency issues. To examine this, we capture information asymmetries surrounding the firm from three different angles. First, we examine analyst coverage as analysts are known to be essential in propagating bad news (Hong et al., 2000), and greater analyst coverage subsequently lowers crash risk (Kim et al., 2019). We expect that with a greater number of analysts following, the impact on crash risk will weaken as information asymmetry is reduced.

The other two measures that we use proxy for opinion divergence. We follow Sadka and Scherbina (2007) and calculate analyst forecast dispersion, $DISPERSION_t$, as the standard deviation of all outstanding current fiscal year earnings forecasts scaled by the absolute value of the median forecast. A greater dispersion of forecasts among analysts signals difficulty in valuing the firm (Sadka & Scherbina, 2007). This uncertainty can indicate greater opacity, so we would expect DBN laws to have a larger effect on crash risk for these firms. As an alternative to analyst forecast dispersion, we follow Cho et al. (2013) and use bid-ask spreads ($SPREAD_t$) to proxy for firm information asymmetry, as this signals investor disagreement with respect to firm value. Again, this information asymmetry can provide a greater opportunity for managers to hide bad news, subsequently increasing crash risk. We follow Fong et al.'s (2017) suggestion of using the average daily percentage bid-ask spread, for a given fiscal year, as an efficient measure to calculate $SPREAD_t$, where the percentage spread is the bid-ask spread scaled by the mid-point of the two quotes for any given day.

The regression results from subsampling our data using the three measures above are reported in Table 8. Firms are placed into either low or high groups based on a sample median split. We find that the coefficients on $DBNL_t$ are only significant when there are a low number of analysts following (Panel A), high analyst forecast dispersion (Panel B) or high bid-ask spreads (Panel C). These results support the argument that the impact of DBN laws on increasing firm stock price crash risk is information environment dependent influencing, in turn, the subsequent implied opportunity for managers to hide bad news.

[19] The PRC (see https://privacyrights.org/data-breaches) is a nonprofit consumer and advocacy organization that collects information on data breaches. The PRC data only starts in 2005 so we use the observed data breaches across industries from the first 3 years of data (i.e., 2005–2007) to proxy for the industries that have high ex-ante risk for 2001–2005.

[20] Our search targets any titles with the words "information," "technology" or "security" in it. Therefore, we also capture hybrid titles, such as, for example, "Chief Information Security Officer."

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 8 Cross-sectional analysis: managers' ability to hoard bad news.

Panel A. Analyst coverage ($ANALYST_t$)						
	$NSKEW_{t+1}$		$DUVOL_{t+1}$		$COUNT_{t+1}$	
	(1)	(2)	(3)	(4)	(5)	(6)
	Low	High	Low	High	Low	High
$DBNL_t$	0.1337***	0.0854	0.0318***	0.0205	0.0871**	0.0862
	(0.05)	(0.07)	(0.01)	(0.01)	(0.03)	(0.06)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,660	17,772	19,660	17,772	19,660	17,772
Adjusted R^2	0.0895	0.0650	0.1184	0.0732	0.0815	0.0380
Panel B. Dispersion in analyst forecasts ($DISPERSION_t$)						
	$NSKEW_{t+1}$		$DUVOL_{t+1}$		$COUNT_{t+1}$	
	(1)	(2)	(3)	(4)	(5)	(6)
	Low	High	Low	High	Low	High
	0.0546	0.1297**	0.0191	0.0283***	0.0469	0.1359***
	(0.07)	(0.05)	(0.01)	(0.01)	(0.04)	(0.03)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,178	15,233	15,178	15,233	15,178	15,233
Adjusted R^2	0.0681	0.0506	0.0735	0.0781	0.0469	0.0527
Panel C. Bid-ask spread ($SPREAD_t$)						
	$NSKEW_{t+1}$		$DUVOL_{t+1}$		$COUNT_{t+1}$	
	(1)	(2)	(3)	(4)	(5)	(6)
	Low	High	Low	High	Low	High
$DBNL_t$	0.0477	0.1475**	0.0148	0.0337***	0.0669	0.0850**
	(0.07)	(0.06)	(0.01)	(0.01)	(0.05)	(0.03)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,743	18,765	18,743	18,765	18,743	18,765
Adjusted R^2	0.0630	0.1148	0.0680	0.1555	0.0449	0.0926

Notes: This table reports regression results after splitting the data into "Low" and "High" groups based on the median of $ANALYST_t$ in Panel A, $DISPERSION_t$ in Panel B, and $SPREAD_t$ in Panel C. All variables are defined in Appendix 2. Standard errors are in parentheses and are two-way clustered by headquarters state and by year.
***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

4.5 | Robustness tests

We conclude our empirical investigation with several sensitivity tests to ensure that our results are robust to variations in statistical and sampling design. First, we reestimate our baseline regressions using a stacked regression

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

approach. As Baker et al. (2022) highlighted, staggered DiD estimates can be biased due to a combination of the staggered treatment timing and possible treatment effect heterogeneity in the data. This issue is particularly problematic when the staggered sample period is long, as with DBN laws, where the adoption period spans 16 years. To address potential treatment effect heterogeneity, we adopt the stacked DiD approach recommended by Baker et al. (2022) and also applied by Cengiz et al. (2019) to mitigate these concerns.

For each enactment event of a DBN law, we assemble a dataset comprising 3 years before and 3 years after the law's implementation year within the respective state.[21] Firms that have not yet received treatment during this 7-year window constitute the control group for that cohort. These cohorts are then aggregated to construct our final sample. Our panel regression model is modified from our baseline model to include cohort-specific fixed effects, which are interacted with time and firm fixed effects to account for any unobserved heterogeneity that could influence our estimates (Baker et al., 2022). To ensure we have sufficient clean controls for each cohort, the last event that is considered is in 2014. The results, displayed in Table 9, demonstrate that the impact of DBN laws on crash risk remains significant even if we only focus on the years directly preceding and succeeding the event itself.

We also conduct a few additional tests that are presented in the Supporting Information Appendix. Throughout the paper, we identify changes in data breach disclosure requirements based on the state where firms are headquartered. However, if firms have a substantial number of employees and/or customers outside of their home state, they will also be exposed to the disclosure requirements of those states. Indeed, firms that conduct most of their sales via the internet will also have a diverse customer-base. This will weaken the effect that we pick up from just using the firm's home state (i.e., where it is headquartered). Although we cannot observe states where a firm operates and has customers, using Compustat, we determine the number of operational segments as a proxy for diversification. We reestimate our model including the variable *SEGMENT* that counts segments and interact it with the DBN law indicator. Our baseline results continue to hold in Table O4 although, as one would expect, firms with multiple segments experience a reduced DBN law effect.

In Table O5 of the Supporting Information Appendix, we report results from exclusion tests and higher order fixed effects tests. In Panel A, we exclude the years 2007 and 2008 to account for the impact of the global financial crisis on crash risk. In Panel B, we include industry-year fixed effects. In both cases, the coefficient of $DBNL_t$ remains positive and significant.

5 | CONCLUSION

As Leuz and Wysocki (2016) highlighted, there is a need for further studies examining the effects of nonfinancial disclosure regulation on corporate behavior. This study examines nonfinancial disclosure rules related to the staggered enactment of DBN laws across states to investigate if they alter the incentives of managers to hoard bad news, and, in turn, if they impact stock price behavior. Our results show that DBN laws incentivize managers to withhold negative information to counter the cost of potentially having to declare data breaches in the future. In addition, we find that the laws have a stronger effect on firms where managers have a greater incentive to hoard bad news or where they have a greater ability and opportunity to hoard information.

Overall, our findings support the notion that DBN laws can substantially affect managerial behavior and have a capital market effect. Ironically, in the case of DBN laws, we find evidence that although the intention of these laws is to provide greater operational transparency, it also incentivizes managers to withhold other negative news which adversely affects firm financial disclosure practices. Nevertheless, not all types of nonfinancial disclosure laws will have this effect, as it will depend on whether legislation directly affects firms' agency problems. This provides an

[21] Although, for completeness, we include Alabama, New Mexico and South Dakota, these states adopt the laws at the end of our sample period when there are a dwindling number of control firms. We later show that our results are unaffected when dropping the DBN law changes in these states from our analyses.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 9 Stacked difference-in-difference (DiD) regressions.

	$NSKEW_{t+1}$ (1)	$DUVOL_{t+1}$ (2)	$COUNT_{t+1}$ (3)
$DBNL_t$	0.1159**	0.0226**	0.1297***
	(0.05)	(0.01)	(0.03)
$NSKEW_t$	−0.1847***	−0.0360***	−0.0873***
	(0.02)	(0.00)	(0.01)
$KURTOSIS_t$	0.0037***	0.0012***	0.0047***
	(0.00)	(0.00)	(0.00)
RET_t	0.8892**	0.2569***	0.8723***
	(0.34)	(0.07)	(0.22)
$STDEV_t$	−3.4931	−0.8430**	−2.1309**
	(2.28)	(0.42)	(0.95)
MTB_t	0.0145***	0.0033***	0.0093***
	(0.00)	(0.00)	(0.00)
$LEVERAGE_t$	−0.1779	−0.0624**	−0.2851**
	(0.15)	(0.03)	(0.14)
ROE_t	0.0673***	0.0138***	0.0294*
	(0.02)	(0.00)	(0.02)
$SIZE_t$	0.1451***	0.0381***	0.1268***
	(0.01)	(0.00)	(0.01)
$DTURN_t$	0.0338***	0.0066***	0.0137
	(0.01)	(0.00)	(0.01)
$OPACITY_t$	0.1338	0.0258	0.1696*
	(0.15)	(0.03)	(0.10)
$RELIGION_t$	0.2179	0.1306	0.0368
	(0.73)	(0.14)	(0.91)
LIT_RISK_t	−0.4040**	−0.0887**	−0.4872***
	(0.19)	(0.04)	(0.17)
$INST_OWN_t$	0.3501***	0.0635**	0.0827
	(0.12)	(0.03)	(0.11)
$ANALYST_t$	0.1684***	0.0335***	0.0708**
	(0.03)	(0.01)	(0.03)
Intercept	−2.4782***	−0.7049***	−1.9552***
	(0.42)	(0.09)	(0.51)
Firm-cohort F.E.	Yes	Yes	Yes
Year-cohort F.E.	Yes	Yes	Yes
Observations	24,307	24,307	24,307
Adjusted R^2	0.1083	0.1329	0.0713

(Continues)

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 9 (Continued)

Notes: This table presents results from stacked DiD analyses. We create cohort-specific datasets comprising of treatment firms in each state 3 years before and after each DBN law change (i.e., a 7-year window) and a set of clean controls (firms in states with no DBN laws during the event window). These datasets are then aligned by their event-time to form a stacked dataset that is utilized for the regression analysis. Cohort-specific fixed effects are included and interacted with the year and firm fixed effects. The last event that is considered is in 2014 to ensure that we have sufficient clean controls for each cohort. All variables are defined in Appendix 2. Standard errors are in parentheses and are two-way clustered by headquarters state and by year.
***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

avenue for future research to consider what type of nonfinancial disclosure laws are more or less prone to have this effect.

ACKNOWLEDGMENTS

We are grateful for the constructive feedback provided by both the editor, Peter Pope, and an anonymous reviewer in shaping the final draft. We also thank Shangyi Gao, Shihe Li and Jinhong Park for research assistance. All errors remain those of the authors.

Open access publishing facilitated by The University of Adelaide, as part of the Wiley - The University of Adelaide agreement via the Council of Australian University Librarians.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from a number of third party providers that are cited in the paper. Restrictions apply to the availability of these data as they require a paid subscription.

REFERENCES

Ablon, L., Heaton, P., Lavery, D. C., & Romanosky, S. (2016). *Consumer attitudes toward data breach notifications and loss of personal information*. Rand Corporation.

An, H., & Zhang, T. (2013). Stock price synchronicity, crash risk, and institutional investors. *Journal of Corporate Finance*, *21*, 1–15.

Amir, E., Levi, S., & Livne, T. (2018). Do firms underreport information on cyber-attacks? Evidence from capital markets. *Review of Accounting Studies*, *23*(3), 1177–1206.

Andreou, P. C., Louca, C., & Petrou, A. P. (2017). CEO age and stock price crash risk. *Review of Finance*, *21*(3), 1287–1325.

Ashraf, M., & Sunder, J. (2023). Can shareholders benefit from consumer protection disclosure mandates? Evidence from data breach disclosure laws. *The Accounting Review*, *98*(4), 1–32.

Baker, A. C., Larcker, D. F., & Wang, C. C. (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics*, *144*(2), 370–395.

Ball, R., Jayaraman, S., & Shivakumar, L. (2012). Audited financial reporting and voluntary disclosure as complements: A test of the confirmation hypothesis. *Journal of Accounting and Economics*, *53*(1–2), 136–166.

Bao, D., Kim, Y., Mian, G. M., & Su, L. (2019). Do managers disclose or withhold bad news? Evidence from short interest. *The Accounting Review*, *94*(3), 1–26.

Beyer, A., Cohen, D. A., Lys, T. Z., & Walther, B. R. (2010). The financial reporting environment: Review of the recent literature. *Journal of Accounting and Economics*, *50*(2), 296–343.

Bird, A., Karolyi, S. A., Ruchti, T. G., & Truong, P. (2021). More is less: Publicizing information and market feedback. *Review of Finance*, *25*(3), 745–775.

Boasiako, K. A., & Keefe, M. O. C. (2021). Data breaches and corporate liquidity management. *European Financial Management*, *27*(3), 528–551.

Bourveau, T., Lou, Y., & Wang, R. (2018). Shareholder litigation and corporate disclosure: Evidence from derivative lawsuits. *Journal of Accounting Research*, *56*(3), 797–842.

Callen, J. L., & Fang, X. (2015a). Religion and stock price crash risk. *Journal of Financial and Quantitative Analysis*, *50*(1–2), 169–195.

Callen, J. L., & Fang, X. (2015b). Short interest and stock price crash risk. *Journal of Banking and Finance*, *60*, 181–194.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Cao, F., Sun, J., & Yuan, R. (2019). Board directors with foreign experience and stock price crash risk: Evidence from China. *Journal of Business Finance and Accounting*, *46*, 1144–1170.

Cengiz, D., Dube, A., Lindner, A., & Zipperer, B. (2019). The effect of minimum wages on low-wage jobs. *The Quarterly Journal of Economics*, *134*(3), 1405–1454.

Chen, J., Henry, E., & Jiang, X. (2023). Is cybersecurity risk factor disclosure informative? Evidence from disclosures following a data breach. *Journal of Business Ethics*, *187*, 199–224.

Chen, J., Hong, H., & Stein, J. C. (2001). Forecasting crashes: Trading volume, past returns, and conditional skewness in stock prices. *Journal of Financial Economics*, *61*(3), 345–381.

Chen, Y. C., Hung, M., & Wang, Y. (2018). The effect of mandatory CSR disclosure on firm profitability and social externalities: Evidence from China. *Journal of Accounting and Economics*, *65*(1), 169–190.

Chen, X., Hilary, G., & Tian, X. S. (2019). *Data breach disclosure and insider trading* [Working paper].

Cho, S. Y., Lee, C., & Pfeiffer, R. J., Jr. (2013). Corporate social responsibility performance and information asymmetry. *Journal of Accounting and Public Policy*, *32*(1), 71–83.

Christensen, H. B., Floyd, E., Liu, L. Y., & Maffett, M. (2017). The real effects of mandated information on social responsibility in financial reports: Evidence from mine-safety records. *Journal of Accounting and Economics*, *64*(2–3), 284–304.

Christophe, S. E., Ferri, M. G., & Angel, J. J. (2004). Short-selling prior to earnings announcements. *The Journal of Finance*, *59*(4), 1845–1876.

Dechow, P. M., Sloan, R. G., & Sweeney, A. P. (1995). Detecting earnings management. *Accounting review*, *70*(2), 193–225.

Desai, H., Ramesh, K., Thiagarajan, S. R., & Balachandran, B. V. (2002). An investigation of the informational role of short interest in the Nasdaq market. *The Journal of Finance*, *57*(5), 2263–2287.

Dimson, E. (1979). Risk measurement when shares are subject to infrequent trading. *Journal of Financial Economics*, *7*(2), 197–226.

Fong, K. Y., Holden, C. W., & Trzcinka, C. A. (2017). What are the best liquidity proxies for global research? *Review of Finance*, *21*(4), 1355–1401.

Friedman, H. L., Hughes, J. S., & Michaeli, B. (2022). A rationale for imperfect reporting standards. *Management Science*, *68*(3), 2028–2046.

Froot, K. A., & Stein, J. C. (1998). Risk management, capital budgeting, and capital structure policy for financial institutions: An integrated approach. *Journal of Financial Economics*, *47*(1), 55–82.

Garg, P. (2020). Cybersecurity breaches and cash holdings: Spillover effect. *Financial Management*, *49*(2), 503–519.

Gennaioli, N., Shleifer, A., & Vishny, R. (2015). Neglected risks: The psychology of financial crises. *The American Economic Review*, *105*(5), 310–314.

Greenstone, M., Oyer, P., & Vissing-Jorgensen, A. (2006). Mandated disclosure, stock returns, and the 1964 Securities Acts amendments. *The Quarterly Journal of Economics*, *121*(2), 399–460.

Hanlon, D., Khedmati, M., Lim, E. K. Y., & Truong, C. (2023). Boardroom backscratching and stock price crash risk. *Journal of Business Finance and Accounting*. https://doi.org/10.1111/jbfa.12748

Healy, P. M., & Palepu, K. G. (2001). Information asymmetry, corporate disclosure, and the capital markets: A review of the empirical disclosure literature. *Journal of Accounting and Economics*, *31*(1), 405–440.

Heinle, M., Samuels, D., & Taylor, D. (2022). Disclosure substitution. *Management Science*, *69*(8), 4774–4789.

Hirst, D. E., Koonce, L., & Venkataraman, S. (2007). How disaggregation enhances the credibility of management earnings forecasts. *Journal of Accounting Research*, *45*(4), 811–837.

Hoberg, G., & Phillips, G. (2010). Product market synergies and competition in mergers and acquisitions: A text-based analysis. *The Review of Financial Studies*, *23*(10), 3773–3811.

Hoberg, G., & Phillips, G. (2016). Text-based network industries and endogenous product differentiation. *Journal of Political Economy*, *124*(5), 1423–1465.

Hong, H. A., Kim, J. B., & Welker, W. (2017). Divergence of cash flow and voting rights, opacity, and stock price crash risk: International evidence. *Journal of Accounting Research*, *55*, 1167–1212.

Hong, H., Lim, T., & Stein, J. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, *55*(1), 265–295.

Huang, H. H., & Wang, C. (2021). Do banks price firms' data breaches? *The Accounting Review*, *96*(3), 261–286.

Hutton, A. P., Marcus, A. J., & Tehranian, H. (2009). Opaque financial reports, R^2, and crash risk. *Journal of Financial Economics*, *94*(1), 67–86.

Jin, L., & Myers, S. C. (2006). R^2 around the world: New theory and new tests. *Journal of Financial Economics*, *79*(2), 257–292.

Kamiya, S., Kang, J.-K., Kim, J., Milidonis, A., & Stulz, R. M. (2021). Risk management, firm reputation, and the impact of successful cyberattacks on target firms. *Journal of Financial Economics*, *139*(3), 719–749.

Karamanou, I., Pownall, G., & Prakash, R. (2021). Asymmetric information consolidation and price discovery: Inferring bad news from insider sales. *Journal of Business Finance and Accounting*, *48*, 230–268.

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Kim, J.-B., Li, Y., & Zhang, L. (2011). Corporate tax avoidance and stock price crash risk: Firm-level analysis. *Journal of Financial Economics*, *100*(3), 639–662.

Kim, J.-B., Lu, L. Y., & Yu, Y. (2019). Analyst coverage and expected crash risk: Evidence from exogenous changes in analyst coverage. *The Accounting Review*, *94*(4), 345–364.

Kothari, S. P., Shu, S., & Wysocki, P. D. (2009). Do managers withhold bad news? *Journal of Accounting Research*, *47*(1), 241–276.

Leuz, C., & Wysocki, P. D. (2016). The economics of disclosure and financial reporting regulation: Evidence and suggestions for future research. *Journal of Accounting Research*, *54*(2), 525–622.

Li, X., & Yang, H. I. (2016). Mandatory financial reporting and voluntary disclosure: The effect of mandatory IFRS adoption on management forecasts. *The Accounting Review*, *91*(3), 933–953.

Liu, J., & Ni, X. (2024). Ordeal by innocence in the big-data era: Intended data breach disclosure, unintended real activities manipulation. *European Financial Management*, *30*, 129–163.

Mian, G. M., & Sankaraguruswamy, S. (2012). Investor sentiment and stock market response to earnings news. *The Accounting Review*, *87*(4), 1357–1384.

Murciano-Goroff, R. (2019). Do data breach disclosure laws increase firms' investment in securing their digital infrastructure? Workshop on the Economics of Information Security. https://weis2019.econinfosec.org/wp-content/uploads/sites/6/2019/05/WEIS_2019_paper_33.pdf

Noh, S., So, E. C., & Weber, J. P. (2019). Voluntary and mandatory disclosures: Do managers view them as substitutes? *Journal of Accounting and Economics*, *68*(1), 101243.

Rosati, P., & Lynn, T. (2021). A dataset for accounting, finance and economics research on U.S. data breaches. *Data in Brief*, *35*, 106924.

Sadka, R., & Scherbina, A. (2007). Analyst disagreement, mispricing, and liquidity. *The Journal of Finance*, *62*(5), 2367–2403.

Winn, J. K. (2009). Are better security breach notification laws possible? *Berkeley Technological Law Journal*, *24*(3), 1133–1165.

Xu, H., Guo, S., Haislip, J. Z., & Pinsker, R. E. (2019). Earnings management in firms with data security breaches. *Journal of Information Systems*, *33*(3), 267–284.

Xu, Y., Xuan, Y., & Zheng, G. (2021). Internet searching and stock price crash risk: Evidence from a quasi-natural experiment. *Journal of Financial Economics*, *141*(1), 255–275.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Obaydin, I., Xu, L., & Zurbruegg, R. (2024). The unintended cost of data breach notification laws: Evidence from managerial bad news hoarding. *Journal of Business Finance & Accounting*, *51*, 2709–2736. https://doi.org/10.1111/jbfa.12794

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

APPENDIX I: ENACTMENT DATES OF DATA BREACH NOTIFICATION LAWS AND PROVISIONS ALLOWING FOR A PRIVATE CAUSE OF ACTION

State	Effective date	State	Effective date	State	Effective date
Alabama*	6/1/2018	Kentucky	7/15/2014	North Dakota*	6/1/2005
Alaska*	1/1/2009	Louisiana*	1/1/2006	Ohio	2/17/2006
Arizona	12/31/2006	Maine	1/31/2006	Oklahoma	11/1/2008
Arkansas	8/12/2005	Maryland*	1/1/2008	Oregon*	10/1/2007
California*	7/1/2003	Massachusetts*	10/31/2007	Pennsylvania	6/20/2006
Colorado	9/1/2006	Michigan	7/2/2007	Rhode Island*	3/1/2006
Connecticut	1/1/2006	Minnesota*	1/1/2006	South Carolina*	7/1/2009
Delaware*	6/28/2005	Mississippi	7/1/2011	South Dakota	7/1/2018
District of Columbia*	7/1/2007	Missouri	8/28/2009	Tennessee*	7/1/2005
Florida	7/1/2014	Montana	3/1/2006	Texas*	4/1/2009
Georgia	5/5/2005	Nebraska	7/14/2006	Utah	1/1/2007
Hawaii*	1/1/2007	Nevada*	1/1/2006	Vermont	5/8/2012
Idaho*	7/1/2006	New Hampshire*	1/1/2007	Virginia	7/1/2008
Illinois*	6/27/2006	New Jersey*	1/1/2006	Washington*	7/24/2005
Indiana	7/1/2006	New Mexico	6/16/2017	West Virginia	6/6/2008
Iowa	7/1/2008	New York*	12/7/2005	Wisconsin*	3/31/2006
Kansas	1/1/2007	North Carolina*	12/1/2005	Wyoming	7/1/2007

Note: This table reports the first time in each state and district that a data breach law is enacted specifically containing security breach notification provisions. For example, Nevada introduced a data breach statute in January 2005 but it was not until January 2006 that it required notification provisions for general data breaches. The "*" indicates which states allow for a private cause of action. Only in Nevada is the ability to launch a private action (1 October 2005) different to when notification laws are first effective in a state. The data are collated from Lexis Advance, I.T. Governance (see www.itgovernanceusa.com) and Perkinscoie (see https://www.perkinscoie.com).

APPENDIX II: VARIABLE DEFINITIONS

Definitions of the variables used in this paper are provided below. Unless otherwise stated, all variables are based on a fiscal year and winsorized at the 1% and 99% percentiles.

Variable name	Description	Source
Variable of interest		
$DBNL_t$	Indicator variable set to one if a firm's headquarters are in a state that has enacted a data breach notification law, and zero otherwise	Lexis Advance, I.T. Governance, Perkinscoie
Stock price crash risk variables		
$COUNT_{t+1}$	The difference in the number of times firm-specific daily returns are above, to those that are below, k standard deviations of the mean during the fiscal year, where k is set to generate frequencies of 0.1% in the log-normal distribution	CRSP

(Continues)

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Variable name	Description	Source
$DUVOL_{t+1}$	Natural log of down-to-up volatility of daily firm-specific stock returns measured over the fiscal year $t+1$. See Equation (4)	CRSP
$NSKEW_{t+1}$	Negative coefficient of skewness of daily firm-specific stock returns during the fiscal year $t+1$. See Equation (3)	CRSP
Other variables		
$ANALYST_t$	The logarithm of one plus the number of analysts covering a firm in the fiscal year	I/B/E/S
$CONVERT_t$	Indicator variable equal to one if the firm has convertible bonds or convertible preferred stocks outstanding and zero otherwise	Compustat
$DISPERSION_t$	The standard deviation of all outstanding current fiscal year analyst earnings forecasts scaled by the absolute value of the mean forecast	I/B/E/S
$DTURN_t$	Average monthly share turnover over the fiscal year t minus average monthly share turnover over the previous fiscal year ($t-1$). Monthly share turnover is defined as monthly share trading volume scaled by the total number of shares outstanding for the month	CRSP
$EARNINGS_VOL_t$	The standard deviation of the annual return on assets of a firm over the previous 10 years	Compustat
$INST_OWN_t$	Proportion of the firm owned by institutional investors at the end of the fiscal year	Thomson 13F
$KURTOSIS_t$	Kurtosis of daily firm-specific stock returns over the fiscal year	CRSP
$LEVERAGE_t$	Total long-term debt to total assets at the end of the fiscal year	Compustat
LIT_RISK_t	Indicator equal to one for firms in the biotechnology, computer, electronics or retail industries, and zero otherwise	Compustat
$LOSS_t$	Indicator variable equal to one if income before extraordinary items of year $t-1$ is negative, and zero otherwise	Compustat
MTB_t	Market to book value of equity at the end of the fiscal year	Compustat
$NEGK_t$	One plus the logarithm of the number of negative voluntary 8K filings (based on CAR reaction surrounding to the event) in a fiscal year	Compustat
$O\text{-}P_DIFF_t$	Difference between $OPTIMISTIC_t$ and $PESSIMISTIC_t$	N/A
$OPACITY_t$	Three years rolling sum of absolute abnormal accruals as estimated from the modified Jones model (Dechow et al., 1995)	Compustat
$OPTIMISTIC_t$	Count of the number of times earnings guidance in the fiscal year is above actual earnings	Compustat

(Continues)

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Variable name	Description	Source
$PESSIMISTIC_t$	Count of the number of times earnings guidance in the fiscal year is below actual earnings	Compustat
$POS8K_t$	One plus the logarithm of the number of positive voluntary 8K filings (based on CAR reaction surrounding to the event) in a fiscal year	Compustat
$RELIGION_t$	Proportion of religious adherents in a state relative to the population	www.thearda.com
$ResSI_t$	Residual short interest calculated from Equation (6)	N/A
RET_t	Cumulative firm-specific daily stock returns over the fiscal year	CRSP
ROE_t	Income before extraordinary items over the book value of equity at the end of the fiscal year	Compustat
SCA_t	Indicator equal to one if the firm has a securities class action lodged against it in the fiscal year, and otherwise zero	Stanford Law School Securities Class Action Clearinghouse filings database (https://securities.stanford.edu/filings.html)
$SEGMENT_t$	The number of operational segments the firm has in the fiscal year	Compustat
SI_t	A firm's short interest at the end of the fiscal year	Compustat
$SIZE_t$	Natural logarithm of the market value of equity at the end of the fiscal year	CRSP
$SPREAD_t$	Average daily percentage bid-ask spread where the percentage spread is the bid-ask spread scaled by the mid-point of the two quotes for any given day	CRSP
$STDEV_t$	Standard deviation of firm-specific daily stock returns over the fiscal year	CRSP
$TREND_t$	An annual time trend variable	N/A
$VOL8K_t$	One plus the logarithm of the number of voluntary 8K filings in a fiscal year	Compustat

14685957, 2024, 9-10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12794 by University Of West Florida, Wiley Online Library on [07/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License